

```
!pip install -q youtube-transcript-api
!pip install -q langchain langchain-community
!pip install -q faiss-cpu
!pip install -q sentence-transformers
!pip install -q langchain-google-genai
!pip install -q google-generativeai
!pip install -q streamlit
!pip install -q langchain-text-splitters
!pip install -qU langchain-huggingface
```

## STEP - 1 : Import Libraries

```
from youtube_transcript_api import YouTubeTranscriptApi,
TranscriptsDisabled
from youtube_transcript_api.formatters import TextFormatter
from langchain_text_splitters import RecursiveCharacterTextSplitter,
Language
from langchain_huggingface import HuggingFaceEmbeddings
from langchain_community.vectorstores import FAISS
from langchain_core.prompts import PromptTemplate
from dotenv import load_dotenv
import os
load_dotenv()

C:\Users\abhis\AppData\Local\Temp\ipykernel_23000\599392212.py:5:
DeprecationWarning: `langchain-community` is being sunset and is no
longer actively maintained. See
https://github.com/langchain-ai/langchain-community/issues/674 for
details and migration guidance toward standalone integration packages.
  from langchain_community.vectorstores import FAISS

False

from langchain_google_genai import ChatGoogleGenerativeAI

import warnings
warnings.filterwarnings('ignore')
```

## STEP - 2 : Connect Gemini

```
from langchain_google_genai import ChatGoogleGenerativeAI

GOOGLE_API_KEY="AQ.Ab8RN6JrEw6P9BSDd8hGiFCPG4rIT8DYk6oMUgyCUFu2ti-M7A"

llm = ChatGoogleGenerativeAI(
    model="gemini-3.6-flash",
    google_api_key=GOOGLE_API_KEY,
    temperature=0.2
)
```

## STEP - 3 : Extract YouTube Transcript

```
video_id = "Gfr50f6ZBvo"

ytt_api = YouTubeTranscriptApi()

transcript_data = ytt_api.fetch(video_id)

transcript = " ".join(
    [item.text for item in transcript_data]
)

print(transcript[:500])
```

the following is a conversation with demus hasabis ceo and co-founder of deepmind a company that has published and builds some of the most incredible artificial intelligence systems in the history of computing including alfred zero that learned all by itself to play the game of go better than any human in the world and alpha fold two that solved protein folding both tasks considered nearly impossible for a very long time demus is widely considered to be one of the most brilliant and impactful

## STEP - 4: Split Transcript into Chunks

```
splitter = RecursiveCharacterTextSplitter(chunk_size=1000,
chunk_overlap=200)
chunks = splitter.create_documents([transcript])

len(chunks)

168
```

## STEP - 5 : Create Embeddings

```
embeddings = HuggingFaceEmbeddings(
    model_name="all-MiniLM-L6-v2"
)
```

Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF\_TOKEN to enable higher rate limits and faster downloads.

```
{"model_id": "0f629ffd96f448efa5fd0baa3f9c1974", "version_major": 2, "version_minor": 0}
```

## STEP - 6 : Create FAISS Vector Store

```
vector_store = FAISS.from_documents(chunks, embeddings)
```

```
print("Vector DB Created")
```

Vector DB Created

## STEP - 7 : Retriever

```
retriever = vector_store.as_retriever(search_type="similarity",  
search_kwargs={"k": 4})
```

retriever

```
VectorStoreRetriever(tags=['FAISS', 'HuggingFaceEmbeddings'],  
vectorstore=<langchain_community.vectorstores.faiss.FAISS object at  
0x0000023F6E4C92B0>, search_kwargs={'k': 4})
```

## STEP - 8 : Create Rag Prompt

```
prompt = PromptTemplate(  
    template="""  
        You are a helpful assistant.  
        Answer ONLY from the provided transcript context.  
        If the context is insufficient, just say you don't know.
```

```
        {context}  
        Question: {question}  
    """,  
    input_variables = ['context', 'question']  
)
```

```
question = "is the topic of nuclear fusion discussed in this  
video? if yes then what was discussed"
```

```
retrieved_docs = retriever.invoke(question)  
retrieved_docs
```

```
[Document(id='336827f2-0fbb-46f4-96ef-e145cb4ec7e5', metadata={},  
page_content="in this case in fusion we we collaborated with epfl in  
switzerland the swiss technical institute who are amazing they have a  
test reactor that they were willing to let us use which you know i  
double checked with the team we were going to use carefully and safely  
i was impressed they managed to persuade them to let us use it and um  
and it's a it's an amazing test reactor they have there and they try  
all sorts of pretty crazy experiments on it and um the the the what we  
tend to look at is if we go into a new domain like fusion what are all  
the bottleneck problems uh like thinking from first principles you  
know what are all the bottleneck problems that are still stopping  
fusion working today and then we look at we you know we get a fusion
```

expert to tell us and then we look at those bottlenecks and we look at the ones which ones are amenable to our ai methods today yes right and and and then and would be interesting from a research perspective from our point of view from an ai point of"),

Document(id='26edefb9-60d8-4d50-8fbd-05ele9427962', metadata={}, page\_content='that i would like to i think could be very transformative if we helped accelerate and uh really interesting problems scientific challenges in of themselves this is energy so energy yes exactly so energy and climate so we talked about disease and biology as being one of the biggest places i think ai can help with i think energy and climate uh is another one so maybe they would be my top two um and fusion is one one area i think ai can help with now fusion has many challenges mostly physics material science and engineering challenges as well to build these massive fusion reactors and contain the plasma and what we try to do whenever we go into a new field to apply our systems is we look for um we talk to domain experts we try and find the best people in the world to collaborate with um in this case in fusion we we collaborated with epfl in switzerland the swiss technical institute who are amazing they have a test reactor that they were willing to let us use which you know i double'),

Document(id='226149ef-fafc-4ab1-a208-ab52ed9a5764', metadata={}, page\_content="so we with this problem and we published it in a nature paper last year uh we held the fusion that we held the plasma in specific shapes so actually it's almost like carving the plasma into different shapes and control and hold it there for the record amount of time so um so that's one of the problems of of fusion sort of um solved so i have a controller that's able to no matter the shape uh contain it continue yeah contain it and hold it in structure and there's different shapes that are better for for the energy productions called droplets and and and so on so um so that was huge and now we're looking we're talking to lots of fusion startups to see what's the next problem we can tackle uh in the fusion area so another fascinating place in a paper title pushing the frontiers of density functionals by solving the fractional electron problem so you're taking on modeling and simulating the quantum mechanical behavior of electrons yes um can you explain this work and can ai model and"),

Document(id='c4c1306f-847a-4a7b-8e57-638216d95ab4', metadata={}, page\_content="like room temperature superconductors or something on my list one day i'd like to like you know have an ai system to help build better optimized batteries all of these sort of mechanical things mr i think a systematic sort of search could be uh guided by a model could be um could be extremely powerful so speaking of which you have a paper on nuclear fusion uh magnetic control of tokamak plasmas to deep reinforcement learning so you uh you're seeking to solve nuclear fusion with deep rl so it's doing control of high temperature plasmas can you explain this work and uh can ai eventually solve nuclear fusion it's been very fun last year or two and very productive because we've been taking off a lot of my dream projects if you like of things that i've collected over the years of areas of science that i would

like to i think could be very transformative if we helped accelerate and uh really interesting problems scientific challenges in of themselves this is energy so energy yes exactly so"]]

```
context_text = "\n\n".join(doc.page_content for doc in retrieved_docs)
final_prompt = prompt.invoke({"context": context_text, "question":
question})
final_prompt
```

```
StringPromptValue(text="\n      You are a helpful assistant.\n
Answer ONLY from the provided transcript context.\n      If the
context is insufficient, just say you don't know.\n\n      in this
case in fusion we we collaborated with epfl in switzerland the swiss
technical institute who are amazing they have a test reactor that they
were willing to let us use which you know i double checked with the
team we were going to use carefully and safely i was impressed they
managed to persuade them to let us use it and um and it's a it's an
amazing test reactor they have there and they try all sorts of pretty
crazy experiments on it and um the the the what we tend to look at is
if we go into a new domain like fusion what are all the bottleneck
problems uh like thinking from first principles you know what are all
the bottleneck problems that are still stopping fusion working today
and then we look at we you know we get a fusion expert to tell us and
then we look at those bottlenecks and we look at the ones which ones
are amenable to our ai methods today yes right and and and then and
would be interesting from a research perspective from our point of
view from an ai point of\n\nthat i would like to i think could be very
transformative if we helped accelerate and uh really interesting
problems scientific challenges in of themselves this is energy so
energy yes exactly so energy and climate so we talked about disease
and biology as being one of the biggest places i think ai can help
with i think energy and climate uh is another one so maybe they would
be my top two um and fusion is one one area i think ai can help with
now fusion has many challenges mostly physics material science and
engineering challenges as well to build these massive fusion reactors
and contain the plasma and what we try to do whenever we go into a new
field to apply our systems is we look for um we talk to domain experts
we try and find the best people in the world to collaborate with um in
this case in fusion we we collaborated with epfl in switzerland the
swiss technical institute who are amazing they have a test reactor
that they were willing to let us use which you know i double\n\nso we
with this problem and we published it in a nature paper last year uh
we held the fusion that we held the plasma in specific shapes so
actually it's almost like carving the plasma into different shapes and
control and hold it there for the record amount of time so um so
that's one of the problems of of fusion sort of um solved so i have a
controller that's able to no matter the shape uh contain it continue
yeah contain it and hold it in structure and there's different shapes
that are better for for the energy productions called droplets and and
and so on so um so that was huge and now we're looking we're talking
```

to lots of fusion startups to see what's the next problem we can tackle uh in the fusion area so another fascinating place in a paper title pushing the frontiers of density functionals by solving the fractional electron problem so you're taking on modeling and simulating the quantum mechanical behavior of electrons yes um can you explain this work and can ai model and\n\nlike room temperature superconductors or something on my list one day i'd like to like you know have an ai system to help build better optimized batteries all of these sort of mechanical things mr i think a systematic sort of search could be uh guided by a model could be um could be extremely powerful so speaking of which you have a paper on nuclear fusion uh magnetic control of tokamak plasmas to deep reinforcement learning so you uh you're seeking to solve nuclear fusion with deep rl so it's doing control of high temperature plasmas can you explain this work and uh can ai eventually solve nuclear fusion it's been very fun last year or two and very productive because we've been taking off a lot of my dream projects if you like of things that i've collected over the years of areas of science that i would like to i think could be very transformative if we helped accelerate and uh really interesting problems scientific challenges in of themselves this is energy so energy yes exactly so\n\nQuestion: is the topic of nuclear fusion discussed in this video? if yes then what was discussed\n\n")

```
answer = llm.invoke(final_prompt)
print(answer.content[0]["text"])
```

Direct use of automatic function calling (AFC) in Models.generate\_content is not recommended. Instead, we recommend to use AFC in Chat.send\_message. Similarly, direct use of AFC in Models.generate\_content\_stream is not recommended. Instead, we recommend to use AFC in Chat.send\_message\_stream.

Yes, the topic of nuclear fusion is discussed.

According to the context, the following points were discussed regarding nuclear fusion:

- \* **Role of AI:** Fusion (under energy and climate) is identified as a major transformative area where AI can help accelerate progress by solving bottleneck problems amenable to current AI methods.
- \* **Challenges:** Fusion faces physics, material science, and engineering challenges, specifically around building massive fusion reactors and containing high-temperature plasma.
- \* **EPFL Collaboration:** The team collaborated with EPFL (the Swiss Technical Institute) in Switzerland to use their test reactor for experiments.
- \* **Deep Reinforcement Learning (Deep RL):** They used deep RL for the magnetic control of tokamak plasmas to control high-temperature plasmas.
- \* **Results / Nature Paper:** They published a paper in Nature

demonstrating a controller that could carve plasma into specific shapes (such as droplets) and contain/hold it for a record amount of time.

**\*\*Next Steps:\*\*** Having solved that plasma containment problem, they are speaking with fusion startups to decide what next problem to tackle in the fusion area.

## STEp - 9 : Building a Chain

```
from langchain_core.runnables import RunnableParallel,  
RunnablePassthrough, RunnableLambda  
from langchain_core.output_parsers import StrOutputParser
```

```
def format_docs(retrieved_docs):  
    context_text = "\n\n".join(doc.page_content for doc in  
retrieved_docs)  
    return context_text
```

```
parallel_chain = RunnableParallel({  
    'context': retriever | RunnableLambda(format_docs),  
    'question': RunnablePassthrough()  
})
```

```
parallel_chain.invoke('who is Demis')
```

```
{'context': "to get world peace because there's also other corrupting  
things like wanting power over people and this kind of stuff which is  
not necessarily satisfied by by just abundance but i think it will  
help um and i think uh but i think ultimately ai is not going to be  
run by any one person or one organization i think it should belong to  
the world belong to humanity um and i think maybe many there'll be  
many ways this will happen and ultimately um everybody should have a  
say in that do you have advice for uh young people in high school and  
college maybe um if they're interested in ai or interested in having a  
big impact on the world what they should do to have a career they can  
be proud of her to have a life they can be proud of i love giving  
talks to the next generation what i say to them is actually two things  
i i think the most important things to learn about and to find out  
about when you're when you're young is what are your true passions is  
first of all there's two things one is find\n\nthe following is a  
conversation with demus hasabis ceo and co-founder of deepmind a  
company that has published and builds some of the most incredible  
artificial intelligence systems in the history of computing including  
alfred zero that learned all by itself to play the game of gold better  
than any human in the world and alpha fold two that solved protein  
folding both tasks considered nearly impossible for a very long time  
demus is widely considered to be one of the most brilliant and  
impactful humans in the history of artificial intelligence and science  
and engineering in general this was truly an honor and a pleasure for  
me to finally sit down with him for this conversation and i'm sure we
```

```
will talk many times again in the future this is the lex friedman
podcast to support it please check out our sponsors in the description
and now dear friends here's demis hassabis let's start with a bit of a
personal question am i an ai program you wrote to interview people
until i get good enough\n\nof deep mind just as a human being let me
ask you about this one particular anecdotal evidence of the google
engineer who made a comment or believed that there's some aspect of a
language model the lambda language model that exhibited sentience so
you said you believe there might be a responsibility to build systems
that are not essential and this experience of a particular engineer i
think i'd love to get your general opinion on this kind of thing but i
think it will happen more and more and more which uh not when
engineers but when people out there that don't have an engineering
background start interacting with increasingly intelligent systems we
anthropomorphize them they they start to have deep impactful um
interactions with us in a way that we miss them yeah when they're gone
and we sure feel like they're living entities self-aware entities and
maybe even we project sentience onto them so what what's your thought
about this particular uh system was is uh have you ever met a\n\nthat
are amazingly smart at certain things like maybe playing go and chess
and other things but they don't feel at all in any shape or form
conscious in the way that you know you do to me or i do to you and um
and i think actually building ai is uh these intelligent constructs uh
is one of the best ways to explore the mystery of consciousness to
break it down because um we're going to have devices that are pretty
smart at certain things or capable of certain things but potentially
won't have any semblance of self-awareness or other things and in fact
i would advocate if there's a choice building systems in the first
place ai systems that are not conscious to begin with uh are just
tools um until we understand them better and the capabilities better
so on that topic just not as the ceo of deep mind just as a human
being let me ask you about this one particular anecdotal evidence of
the google engineer who made a comment or believed that there's some
aspect of a language model the",
'question': 'who is Demis'}
```

```
parser = StrOutputParser()

main_chain = parallel_chain | prompt | llm | parser
```

## STEP - 10 : Genertate Summary

```
main_chain.invoke('summarize the video and also give me the key
points')
```

```
'Based on the provided transcript, here is a summary and the key
points of the conversation:\n\n### **Summary**\n\nThe transcript is an
excerpt from a podcast conversation hosted by Lex with Demis Hassabis
("damas"). They discuss the multi-faceted nature of solving
intelligence at DeepMind—debating the roles of science, engineering,
```



compute, human infrastructure, and philosophy. They touch upon how future fundamental explanations of the world might transcend current models of physics to explain mysteries like consciousness, life, and gravity. Additionally, the conversation reflects on the history of AI and chess, specifically mentioning early efforts by Claude Shannon and Alan Turing, as well as Kasparov's match against Deep Blue.

---

### \*\*Key Points\*\*

- Solving Intelligence:** The challenge of solving intelligence involves a combination of factors, including science, engineering, algorithms, data, hardware/software infrastructure, human collaboration, and philosophy.
- Future Explanations of Physics and Mysteries:** Looking far into the future, fundamental explanations might offer a much deeper or simpler understanding than today's standard model of physics. Such explanations could eventually encompass long-standing mysteries such as consciousness, life, and gravity.
- Historical Milestones in AI and Chess:**
  - Claude Shannon:** Authored an early paper on a chess program (1949).
  - Alan Turing:** Wrote a chess program when computers were too slow to run it, forcing him to act as the computer himself by manually running the program with pencil and paper over two to three days to play a friend.
  - Deep Blue vs. Kasparov:** While Deep Blue's victory was a major milestone, the speaker was more impressed by Kasparov's human mind competing at that level against a brute-force calculation machine.
- Closing:** Lex thanks Demis Hassabis for taking the time to have the conversation and leaves the audience with a partial quote from Edsger Dijkstra: "Computer science is no more about computers than..."

## STEP - 11: Create Questions and Answers

```
while True:

    question = input("Ask a question (type 'exit' to quit): ")

    if question.lower() == "exit":
        break

    # Retrieve relevant documents
    retrieved_docs = retriever.invoke(question)

    # Combine retrieved text
    context_text = "\n\n".join(
        doc.page_content for doc in retrieved_docs
    )

    # Create final prompt
    final_prompt = prompt.invoke({
        "context": context_text,
        "question": question
    })
```

```
# Generate answer using Gemini
answer = llm.invoke(final_prompt)

print("\nAnswer:")
print(answer.content[0]["text"])
print("-" * 80)
```

Ask a question (type 'exit' to quit): exit

## STEP - 12 : Create Quiz

```
main_chain.invoke(
    """
    Create 3 MCQs from the video.
    Provide answers also.
    """
)

'Here are 3 multiple-choice questions based on the provided text,
along with their answers:\n\n---\n\n### **Question 1**\nAccording to
the transcript, what project evolution is used to illustrate the
"AlphaX" methodology?\n* **A)** AlphaFold to AlphaZero\n* **B)**
AlphaGo to AlphaZero\n* **C)** AlphaStar to AlphaCode\n* **D)**
AlphaGo to AlphaStar\n\n**Answer:** **B) AlphaGo to AlphaZero** \
\n*(Context: "...the easiest way to see that is the evolution of
alphago to alpha zero...")*\n\n---\n\n### **Question 2**\nHow long
does it normally take human games testers to balance a game like
Starcraft?\n* **A)** Hundreds of hours\n* **B)** Tens of millions of
hours\n* **C)** Thousands of hours\n* **D)** A few days\n\n**Answer:**
**C) Thousands of hours** \n*(Context: "...which usually takes
thousands of hours from hundreds of games human games testers normally
to to balance...")*\n\n---\n\n### **Question 3**\nWhich computer
scientist\'s quote is referenced at the very end of the excerpt?\n*
**A)** Edsger Dijkstra\n* **B)** Alan Turing\n* **C)** Demis Hassabis\
\n* **D)** Lex Fridman\n\n**Answer:** **A) Edsger Dijkstra** \
\n*(Context: "...let me leave you with some words from edskar dykstra
computer science is no more about computers than...")*'
```